

Batman MCQs — OCR Text Consolidated

OCR transcription of the uploaded image-based PDF. The filename says "100 MCQs", but the document visibly contains 97 MCQs: 27 Easy, 55 Medium, and 15 Difficult. No missing questions were invented. Where the source image cuts off an option, only the visible text is included.

DIFFICULTY: EASY

1. A trainee installs Java to compile and run a program. Which component contains the Java compiler (javac)?

- A. JDK
- B. JRE
- C. JVM only
- D. Garbage Collector

Correct Answer: A. JDK

2. After a Java source file is compiled successfully, what does the compiler normally produce for the JVM to execute? A Anative .exe file

- B. Java bytecode in a .class file C.ASQL script
- D. A compressed JAR only

Correct Answer: B. Java bytecode in a .class file C.ASQL script

3. Car extends Vehicle. What relationship does this represent?

- A. Car has-a Vehicle
- B. Car is-a Vehicle
- C. Vehicle is a method of Car
- D. Car and Vehicle are unrelated

Correct Answer: B. Car is-a Vehicle

4. A class contains print(int value) and print(String value). What concept does this demonstrate?

- A. Overriding
- B. Overloading
- C. Abstraction only
- D. Constructor chaining

Correct Answer: B. Overloading

5. A program must read a plain toxt file containing customer names. Which type of stream is generally more suitable for text?

- A. Character stream
- B. Byte stream only
- C. Object lock
- D. Thread pool

Correct Answer: A. Character stream

6. What is the main benefit of try-with-resources when working with files or JDBC resources?

- A. It automatically closes eligible resources
- B. It makes every exception checked
- C. It stores files in memory
- D. It prevents all runtime errors

Correct Answer: A. It automatically closes eligible resources

7. Which Java class is appropriate for representing a calendar date without a time-of-day or timezone?

- A. LocalDate**
- B. LocalTimeZone
- C. FileDate
- D. ThreadDate

Correct Answer: A. LocalDate

8. Which Scala keyword creates a reference that cannot be reassigned after initialization?

- A. val
- B. var**
- C. def
- D. new

Correct Answer: B. var

9. What is the typical behavior of Scala immutable List operations such as map? A They modify the original list in place

- B. They return a new collection value
- C. They always return Unit
- D. They require a database

Correct Answer: A

10. A Scala program needs to return a customer ID and customer name together without defining a new class. Which simple structure can hold both values? A Tuple

- B. Thread
- C. Exception
- D. RDD action

Correct Answer: A

11. Which of the following is an RDD transformation that returns a new RDD?

- A. map**
- B. count
- C. collect
- D. first

Correct Answer: A. map

12. Which RDD operation is an action that returns the number of elements?

- A. count
- B. filter**
- C. map
- D. flatMap

Correct Answer: B. filter

13. What does lazy evaluation mean for Spark RDD transformations?

- A. They run immediately when written**
- B. Spark records transformations and executes them when an action requires results,
- C. They never execute unless the driver can use them

Correct Answer: A. They run immediately when written

14. Which Scala utility is commonly used in simple examples to read text from

- A. scala.io.Source**
- B. scala.collection.Set
- C. SparkSession.stop
- D. ThreadLocal

Correct Answer: A. scala.io.Source

15. What does Option help represent in Scala?

- A. A value that may be present or absent**
- B. Only a mutable list
- C. A network socket
- D. A thread lock

Correct Answer: A. A value that may be present or absent

16. A file-reading operation may throw an exception. Which Scala type can capture success or failure as a value? A.Try

- B. Tuple2 only :
- C. Array
- D. RDD partition

Correct Answer: A

17. Which DataFrame method is commonly used to inspect how Spark plans to execute a query? 2

- A. explain()**
- B. showSchemaOnly()
- C. collectPlan()
- D. savePlan()

Correct Answer: A. explain()

18. When a transformation can be expressed using built-in Spark SQL functions, which approach is generally preferred?

- A. Use built-in functions**
- B. Always write a custom UDF
- C. Collect data to the driver
- D. Convert everything to text files first

Correct Answer: A. Use built-in functions

19. Why is Parquet often efficient for analytical queries that select only a few columns?

- A. It is columnar and can avoid reading unnecessary columns**
- B. It stores every row as an image
- C. It requires collect() first
- D. It runs only in memory

Correct Answer: A. It is columnar and can avoid reading unnecessary columns

20. Which collection operation transforms every element and returns a new collection? A.map

- B. filter
- C. exists
- D. foreach only

Correct Answer: A

21. Which Scala collection operation keeps only elements that satisfy a Boolean condition?

- A. filter**
- B. map
- C. flatten only
- D. zip only

Correct Answer: A. filter

22. When each input element can produce zero or more output elements and the final result should be flat, which operation is especially useful?

- A. flatMap**
- B. head
- C. size
- D. reverse only

Correct Answer: A. flatMap

23. A CSV contract already defines exact column names and types. What should Spark use when creating the DataFrame for consistent interpretation?

- A. An explicit schema**
- B. A random schema each run
- C. Only a local List
- D. No column names

Correct Answer: A. An explicit schema

24. For structured rows with named columns, filters, joins, and aggregations, which Spark abstraction is usually more suitable?

- A. DataFrame**
- B. Plain focal Array only
- C. Driver-side HashMap only
- D. Thread

Correct Answer: A. DataFrame

25. What does inferSchema do when reading supported data such as CSV?

- A. Spark examines data to guess column data types**
- B. it encrypts the file
- C. It caches all rows permanently
- D. it creates @ JDBC connection

Correct Answer: A. Spark examines data to guess column data types

26. Which Spark API provides DataFrame/Dataset-style processing for streaming data?

- A. Structured Streaming**
- B. Java Senalization C RDD collect
- D. JDBC only

Correct Answer: A. Structured Streaming

27. Which statement best describes DStreams in modern Spark training?

- A. They are the older RDD-based streaming abstraction**
- B. They are Parquet files

Correct Answer: A. They are the older RDD-based streaming abstraction

DIFFICULTY: MEDIUM

1. A training app must store unique course codes and should automatically prevent duplicates. Which collection type best matches this requirement?

- A. List
- B. Set**
- C. Queue
- D. Array

Correct Answer: B. Set

2. A Java method creates a local int variable and also creates a new Customer object. In the usual memory model, where is the object primarily allocated?

- A. Stack only
- B. Heap**
- C. CPU register only
- D. Source-code area

Correct Answer: B. Heap

3. A method should accept a List of String values and prevent callers from accidentally adding Integer values at compile time. Which declaration is most suitable?

- A. List values
- B. List values
- C. List<String> values**
- D. Object values

Correct Answer: C. List<String> values

4. A List receives the primitive value 25 using list.add(26). Why is this allowed?

- A. Java converts the List into an int array
- B. Autoboxing converts int to Integer**
- C. The compiler ignores the type
- D. Integer is a primitive type

Correct Answer: B. Autoboxing converts int to Integer

5. A HashMap stores employee id using employeeId as the key. What happens when put() is called again with the same key and a new value?

- A. A duplicate key is stored
- B. The existing value for that key is replaced**
- C. The map becomes a Set
- D. The key is automatically deleted

Correct Answer: B. The existing value for that key is replaced

6. A trainee must remove selected elements while iterating through an ArrayList and wants to avoid unsafe structural modification, Which approach is appropriate?

- A. Call list.remove() from every foreach iteration
- B. Use Iterator and its remove() method after next()**
- C. Convert the list to a String first
- D. Change the list reference to null

Correct Answer: B. Use Iterator and its remove() method after next()

7. A BankAccount class keeps balance private and exposes deposit() and withdraw() methods. Which OOP principle is being applied?

- A. Encapsulation**
- B. Inheritance
- C. Method overloading
- D. Serialization

Correct Answer: A. Encapsulation

8. A parent class defines calculateFee(), and a child class provides its own implementation with the same signature. Which concept is used?

- A. Method overriding**
- B. Method overloading
- C. Autoboxing
- D. Serialization

Correct Answer: A. Method overriding

9. A child constructor must initialize fields defined by its parent class using the parent constructor. Which keyword is normally used?

- A. this
- B. super**
- C. final
- D. static

Correct Answer: B. super

10. Several unrelated classes must guarantee that they provide a process() method, while each class can implement it differently, Which Java construct best expresses this contract?

- A. interface**
- B. Private constructor
- C. Array
- D. Package only

Correct Answer: A. interface

11. An OrderService uses a PaymentGateway object to process payments, but OrderService is not a type of PaymentGateway. Which design relationship is most appropriate?

- A. Inheritance
- B. Composition**
- C. Overloading
- D. Casting

Correct Answer: B. Composition

12. inside # subclass, which expression refers to the current object, and which refers to the Parent-class implementation?

- A. this and super**
- B. super and this
- C. static and final
- D. new and class

Correct Answer: A. this and super

13. A program generates a simple text report containing formatted lines. Which class is convenient for writing textual output with print/println methods?

- A. PrintWriter**
- B. FileInputStream
- C. ObjectInputStream
- D. HashMap

Correct Answer: A. PrintWriter

14. A trainee wants to check whether a product code follows a known text pattern. Which Java API is intended for regular-expression matching?

- A. Pattern and Matcher**
- B. LocalDate and Period
- C. List and Set
- D. Thread and Runnable only

Correct Answer: A. Pattern and Matcher

15. A List should produce a new list containing only values greater than 100. Which Streams API operation performs the selection?

- A. filter**
- B. map only
- C. reduce only
- D. sorted only

Correct Answer: A. filter

16. A program needs to visit files inside a directory tree using modern Java NIO APIs. Which operation is designed to stream paths recursively?

- A. Files.walk(...)**
- B. System.get()
- C. Math.round(...)
- D. Thread.sleep(...)

Correct Answer: A. Files.walk(...)

17. A Serializable User object contains a temporary sessionToken that should not be written during default Java serialization. Which keyword can mark that field?

- A. transient**
- B. volatile
- C. abstract
- D. extends

Correct Answer: A. transient

18. Scala sees val total = 126.50. What does type inference mean in this statement?

- A. The programmer must write the type later
- B. Scala determines an appropriate type from the assigned value**
- C. The value has no type
- D. The variable becomes a String automatically

Correct Answer: B. Scala determines an appropriate type from the assigned value

19. A status code can be NEW, PAID, or CANCELLED and each value needs a different result. Which Scala feature provides clear case-based branching?

- A. match with case clauses**
- B. Only while loops
- C. SDBC commit
- D. RDD cache

Correct Answer: A. match with case clauses

20. A List(2,4,6) should become List(4,8,12) without changing the original list. Which operation is most suitable? A.map

- B. filter
- C. foreach only
- D. dropWhile only

Correct Answer: A

21. A Scala program needs an indexed collection whose elements can be updated by Position. Which basic collection is appropriate?

- A. Array**
- B. Immutable Tuple only
- C. Option
- D. Try

Correct Answer: A. Array

22. Which expression is @ valid simple Scala lambda that doubles an integer x? A.xe> x2 B.x=x* 2only ee a REST B.x=x* 2only

- C. double(x) class
- D. case object x * 2

Correct Answer: A

23. Why is map usually described as a narrow transformation?

- A. Each output partition can be computed from a limited input partition without a full data shuffle**
- B. It always reduces partitions to one
- C. It broadcasts every record
- D. ttruns only on the driver

Correct Answer: A. Each output partition can be computed from a limited input partition without a full data shuffle

24. A Source is opened to read a text file. What should the program ensure after reading is complete?

- A. The Source is closed**
- B. The JVM is always restarted
- C. The file is converted into an RDD.
- D. The path is stored in a Set

Correct Answer: A. The Source is closed

25. A validation function should return either an error message or a valid Order. Which Scala type naturally models two alternative result types?

- A. Either**
- B. Array only
- C. Unit
- D. Thread

Correct Answer: A. Either

26. A trainee opens `Source.fromFile(path)`. Which method is commonly used to iterate through file lines?

- A. `getLines()`**
- B. `collect()`
- C. `reduceByKey()`
- D. `commit()`

Correct Answer: A. `getLines()`

27. A `Try[Int]` should return 0 when parsing fails. Which concept is designed to provide a fallback for a failed `Try`?

- A. `recover` or `getOrElse`**
- B. RDD checkpoint only
- C. Thread interrupt
- D. `Pattern.compile` only

Correct Answer: A. `recover` or `getOrElse`

28. A Map lookup returns `Option[String]`. Which approach safely provides “Unknown” when no value exists?

- A. `option.getOrElse(“Unknown”)`**
- B. `option.get` always
- C. `option.close()`
- D. `option.collectAsMap()`

Correct Answer: A. `option.getOrElse(“Unknown”)`

29. A Spark job needs to read a table from a relational database. Which Spark data source is appropriate?

- A. JDBC**
- B. Socket text only
- C. RDD checkpoint
- D. `LocalDate`

Correct Answer: A. JDBC

30. A trainee wants to know which join operator Spark actually selected after planning. Which plan is most relevant?

- A. Physical plan**
- B. Only the source file name
- C. DataFrame schema only
- D. Java heap dump

Correct Answer: A. Physical plan

31. A Parquet dataset is partitioned by `event_date`. A query filters `event_date` to one day. What optimization can avoid scanning unrelated date folders?

- A. Partition pruning**
- B. Autoboxing
- C. Thread synchronization
- D. Object serialization

Correct Answer: A. Partition pruning

32. A Parquet table has 160 columns, but a report selects only 6. Which optimization reduces unnecessary column VO?

- A. Column pruning**
- B. Deadlock detection
- C. JDBC rollback
- D. Scala pattern guards

Correct Answer: A. Column pruning

33. A very large JDBC table is currently read through a single input partition. At a conceptual level, what can improve source-side parallel reading when a suitable numeric key exists?

- A. Configure partitioned JDBC reading using a partition column and multiple partitions
- B. Call collect() immediately
- C. Increase only the number of DataFrame columns
- D. Write the table to a Java array first

Correct Answer: A. Configure partitioned JDBC reading using a partition column and multiple partitions

34. A list of transactions must first remove invalid records and then normalize the remaining records. Which chain best matches the required order?

- A. transactions. filter(valid).map(normalize)
- B. transactions.map(normalize).foreach(valid)
- C. transactions foreach(valid) filter(normalize)
- D. transactions.reduce(valid).map(normalize)

Correct Answer: A. transactions. filter(valid).map(normalize)

35. A reusable select function should accept the rule to apply from the caller. Which parameter type represents a predicate for elements of type A?

- A. A=> Boolean
- B. String only
- C. Boolean => A always
- D. Unit => Unit only

Correct Answer: A. A=> Boolean

36. A tax function should return the same result for the same transaction and tax rate every time. Which design best supports this?

- A. Pass required values as parameters and return the result without changing shared state
- B. Read a changing global tax rate
- C. Update a shared mutable variable during calculation
- D. Write to a global buffer before returning

Correct Answer: A. Pass required values as parameters and return the result without changing shared state

37. A method receives an immutable List and must remove invalid values. What should it

- A. Return a new filtered list
- B. Cast and mutate the original list
- C. Store it globally and modify it later
- D. Return Unit and change external state

Correct Answer: A. Return a new filtered list

38. Only active customers should contribute their verified contacts, and each customer can have many contacts. Which chain best fits?

- A. customers filter(_.active).flatMap(_.contacts).filter(_,verified)
- B. customers.map(_.contacts) only
- C. customers foreach(_.contacts).map(_.verified)
- D. customers.head.contacts only

Correct Answer: A. customers filter(_.active).flatMap(_.contacts).filter(_,verified)

39. Two DataFrames contain the same fields in different column orders. Which operation can append rows by matching column names rather than positions?

- A. unionByName
- B. join only
- C. collect
- D. groupByKey on an RDD

Correct Answer: A. unionByName

40. df2 is missing one column that exists in df1. Which option allows unionByName to fill the missing column with null?

- A. allowMissingColumns = true
- B. inferSchema = false
- C. header = true
- D. cache = true

Correct Answer: A. allowMissingColumns = true

41. Two versions produce the same output. One uses DataFrame column expressions; the other converts rows to arbitrary objects and opaque functions. Which version exposes more relational information to Spark?

- A. The DataFrame expression version
- B. The local object version
- C. Both always expose identical semantics
- D. The version that calls collect() first

Correct Answer: A. The DataFrame expression version

42. A large DataFrame result must remain distributed and be written to shared storage. Which approach is appropriate?

- A. Use DataFrameWriter to write the distributed result
- B. collect() everything and use one local file writer
- C. Convert it to one Scala List first
- D. Print every row on the driver

Correct Answer: A. Use DataFrameWriter to write the distributed result

43. A schema declares amount as Decimal, but a source record contains invalid text. Which 'statement is most accurate? .

- A. The schema states the intended type, but malformed input still depends on parsing and validation behavior
- B. The schema guarantees all business values are valid
- C. Spark always repairs every malformed value
- D. Invalid text automatically becomes a valid decimal

Correct Answer: A. The schema states the intended type, but malformed input still depends on parsing and validation behavior

44. A trainer wants a simple live demo where trainees type text into a TCP socket and see Spark process it. Is a socket source reasonable for this purpose?

- A. Yes, for demonstration/testing
- B. No, sockets cannot carry text
- C. Only if the data is first written to JDBC
- D. Only for batch jobs

Correct Answer: A. Yes, for demonstration/testing

45. New CSV files are dropped into a watched folder after they are completely written. Which Structured Streaming source can process newly arriving files?

- A. File source with readStream and a defined schema**
- B. collect() on a static List
- C. Java FileOutputStream only
- D. LocalDate API

Correct Answer: A. File source with readStream and a defined schema

46. A business wants sales totals for each 10-minute interval based on event timestamps. Which concept is appropriate?

- A. Event-time window aggregation**
- B. JVM heap only
- C. Method overloading
- D. Object serialization

Correct Answer: A. Event-time window aggregation

47. Events may arrive several minutes late, but Spark should eventually stop retaining old -event-time state. Which feature addresses accepted lateness and state cleanup?

- A. Watermark**
- B. Autoboxing
- C. JDBC commit
- D. RDD parallelize

Correct Answer: A. Watermark

48. A streaming query restarts after a driver failure. Which design is important if it should continue using previously recorded progress?

- A. Restart with the same durable checkpoint location when the query is compatible**
- B. Use only executor cache
- C. Always create a brand-new checkpoint
- D. Store offsets only in a local mutable variable

Correct Answer: A. Restart with the same durable checkpoint location when the query is compatible

49. An upstream process writes large files slowly. How can it reduce the chance that Spark partially written input files in a watched folder?

- A. Write elsewhere and atomically move/rename completed files into the watched location**
- B. Let Spark read while files are still being written
- C. Delete files before they finish
- D. Store filenames only in driver memory

Correct Answer: A. Write elsewhere and atomically move/rename completed files into the watched location

50. Creating an order and reducing inventory must either both succeed or both fail, Which JDBC transaction approach is appropriate?

- A. Disable auto-commit, commit after both succeed, rollback on failure**
- B. Commit after the first statement regardless of later errors
- C. Use a different connection for each statement with no coordination
- D. Call collect() after each SQL statement

Correct Answer: A. Disable auto-commit, commit after both succeed, rollback on failure

51. Two fields must be updated together so readers never observe a half-updated state. What concept can protect the compound operation?

- A. Use the same shared lock/synchronization around related updates and reads
- B. Use two unrelated local locks created per call . Mark only one field static
- D. Use HashSet instead of synchronization

Correct Answer: A. Use the same shared lock/synchronization around related updates and reads

52. Many threads need to read and update a shared map. Which Java collection is designed for concurrent access better than a plain HashMap without external synchronization?

- A. ConcurrentHashMap
- B. ArrayList
- C. TreeSet only
- D. StringBuilder

Correct Answer: A. ConcurrentHashMap

53. A service wants to represent a result that will be completed asynchronously and then apply another non-blocking step. Which Java API is designed for this style?

- A. CompletableFuture
- B. FileInputStream
- C. LocalDate
- D. HashSet

Correct Answer: A. CompletableFuture

54. A listener list is read very frequently and modified only occasionally. Which concurrent list is known for snapshot-style iteration with relatively expensive writes?

- A. CopyOnWriteArrayList
- B. LinkedHashMap
- C. ArrayDeque
- D. Stringl)

Correct Answer: A. CopyOnWriteArrayList

55. Why do server applications commonly use a JDBC connection pool?

- A. To reuse a limited set of database connections instead of opening a new physical connection for every request
- B. To convert SQL into Scala
- C. To be —_ —_ is fast

Correct Answer: A. To reuse a limited set of database connections instead of opening a new physical connection for every request

DIFFICULTY: DIFFICULT

1. A condition is written as `user != null && user.isActive()`. If `user` is null, why does the expression avoid a `NullPointerException` from `isActive()`?

- A. `&&` evaluates the right side first
- B. `&&` skips the right side when the left side is false**
- C. `isActive()` is converted to static
- D. null is treated as an active user

Correct Answer: B. `&&` skips the right side when the left side is false

2. `Vehicle v = new Car();` is followed by `v.move()`, and `Car` overrides `move()`. Which implementation executes?

- A. `Vehicle.move()` always
- B. `Car.move()` because overridden instance methods use runtime dispatch**
- C. Both methods execute automatically
- D. The code cannot compile a

Correct Answer: B. `Car.move()` because overridden instance methods use runtime dispatch

3. A log line is "ERROR: payment failed". The regex "payment" should detect that word anywhere inside the line. Which Matcher method is the 'better' fit? A `matches()`, because it requires the whole line to match

- B. `find`. because it searches for a matching subsequence**
- C. `replaceAll()`, because it returns Boolean
- D. `group()`, before any match attempt

Correct Answer: B. `find`. because it searches for a matching subsequence

4. A match case should classify only positive amounts above 1000 as "High". Which form best expresses this additional condition?

- A. `case x if x > 1000 => "High"`**
- B. `case x = 1000 => "High"`
- C. `if. case x then 1000`
- D. `case => x > 1000 only`

Correct Answer: A. `case x if x > 1000 => "High"`

5. An RDD partition is lost after an executor failure. How can Spark often rebuild it?

- A. Using the recorded lineage of transformations**
- B. Only from @ manual backup file
- C. By restarting the operating system
- D. By converting it to @ Java List

Correct Answer: A. Using the recorded lineage of transformations

6. A pair RDD contains (product, amount) records. Which operation is commonly used to combine amounts for the same product efficiently?

- A. `reduceByKey`**
- B. `collect` before aggregation
- C. `mapPartitions` only D a to one — _ ae

Correct Answer: A. `reduceByKey`

7. A small Scala collection already exists on the driver for a training example. Which Spark method can distribute it as an RDD?

- A. `sc.parallelize(collection)`
- B. `sc.collect(collection)`
- C. `df.show(collection)`
- D. `spark.stop(collection)`

Correct Answer: A. `sc.parallelize(collection)`

8. The same expensive RDD is reused in several actions during one application. Which 'Spark feature can avoid recomputing it each time?

- A. `persist/cache`
- B. collect then delete the RDD
- C. Rename the RDD
- D. Convert every element to String

Correct Answer: A. `persist/cache`

9. A 500 GB RDD is processed successfully, but the driver fails when `collect()` is called. What is the main reason?

- A. `collect()` attempts to bring all elements to the driver
- B. `collect()` deletes executor memory
- C. `collect()` disables Spark
- D. `collect()` changes every value to null

Correct Answer: A. `collect()` attempts to bring all elements to the driver

10. A program must open a file path and then parse an integer from its contents, and either 'step may fail. Why can wrapping the operations in Try be useful?

- A. It lets success/failure flow through composable operations instead of requiring unchecked assumptions
- B. It guarantees the file exists
- C. It converts every exception to null
- D. It makes all I/O asynchronous

Correct Answer: A. It lets success/failure flow through composable operations instead of requiring unchecked assumptions

11. Two implementations return the same cleaned customer name. One uses built-in trim/upper functions; the other uses a custom UDF. Why is the built-in expression usually easier for Spark to optimize?

- A. its transformation semantics are visible to Spark SQL. planning
- B. It always uses one partition
- C. NM automatically caches all results
- D. It executes only on the driver

Correct Answer: A. its transformation semantics are visible to Spark SQL. planning

12. A pipeline must validate records, convert valid ones into objects, and emit zero or more results per object without mutable buffers. Which sequence best expresses the flow?

- A. filter followed by map followed by flatMap
- B. foreach followed by global ArrayBuffer updates
- C. collect everything first, then mutate a shared list
- D. Use only while loops with shared state

Correct Answer: A. filter followed by map followed by flatMap

13. A workload mainly performs projections, filters, joins, and aggregations on structured order records. The team wants Spark to inspect these operations for optimization. Why is a DataFrame stronger fit than an RDD of opaque domain objects?

- A. DataFrames express named-column relational operations that Spark SQL can analyze
- B. RDDs cannot be distributed
- C. DataFrames always avoid shuffles
- D. RDDs execute only on the driver

Correct Answer: A. DataFrames express named-column relational operations that Spark SQL can analyze

14. A query uses event-time windows and allows events up to 10 minutes late. Why is a 5-minute processing trigger not a replacement for a 10-minute watermark?

- A. The trigger controls how often processing runs, while the watermark relates to event-time lateness/state
- B. The trigger changes event timestamps
- C. The watermark only controls executor count
- D. They are identical settings

Correct Answer: A. The trigger controls how often processing runs, while the watermark relates to event-time lateness/state

15. Transfers can lock two Account objects. One thread locks account 10 then 20, while another may lock 20 then 10. Which simple design reduces deadlock risk while keeping per-account locks?

- A. Acquire required account locks in a consistent deterministic order such as ascending account ID
- B. Always lock the source first regardless of ID
- C. Create a brand-new lock object for each transfer
- D. Hold the first lock while repeatedly sleeping forever

Correct Answer: A. Acquire required account locks in a consistent deterministic order such as ascending account ID

Answer Key

Easy: 1-A, 2-B, 3-B, 4-B, 5-A, 6-A, 7-A, 8-B, 9-A, 10-A, 11-A, 12-B, 13-A, 14-A, 15-A, 16-A, 17-A, 18-A, 19-A, 20-A, 21-A, 22-A, 23-A, 24-A, 25-A, 26-A, 27-A

Medium: 1-B, 2-B, 3-C, 4-B, 5-B, 6-B, 7-A, 8-A, 9-B, 10-A, 11-B, 12-A, 13-A, 14-A, 15-A, 16-A, 17-A, 18-B, 19-A, 20-A, 21-A, 22-A, 23-A, 24-A, 25-A, 26-A, 27-A, 28-A, 29-A, 30-A, 31-A, 32-A, 33-A, 34-A, 35-A, 36-A, 37-A, 38-A, 39-A, 40-A, 41-A, 42-A, 43-A, 44-A, 45-A, 46-A, 47-A, 48-A, 49-A, 50-A, 51-A, 52-A, 53-A, 54-A, 55-A

Difficult: 1-B, 2-B, 3-B, 4-A, 5-A, 6-A, 7-A, 8-A, 9-A, 10-A, 11-A, 12-A, 13-A, 14-A, 15-A